CITIZENS, ACCOUNTABILITY, AND SERVICE SATISFACTION
The Influence of Expectations

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Understanding how citizen-consumers form evaluations of public services is critical to understanding accountability in democratic governance. The task of using citizens' assessments of service quality as an accountability mechanism, however, may be more complex than is commonly understood. In particular, little research has examined how citizens' expectations about the quality of services may influence their levels of satisfaction with public services. In this article, we examine empirically the relationship between perceived performance, expectations, and satisfaction. We examine these relationships across three service areas: trash, police, and schools, relying on survey data from a statewide survey of Georgia residents. Our results suggest higher subjective assessments of service quality are positively related to satisfaction. They also suggest, however, that an "A" service is not always associated with the same level of satisfaction; holding citizens' assessments of service quality constant, positive disconfirmation of expectations increases citizen-consumers' levels of satisfaction with services.

Keywords: *service satisfaction; disconfirmation of expectations; perceived service quality; local service delivery*

Citizens' evaluations of their experiences with service providers provide valuable feedback to agency bureaucrats and to politicians about the effectiveness of the services received. This type of feedback has become perhaps even more critical in light of the reform movement that Osborne and Gaebler (1992) call "reinventing government." While this movement encompasses a broad range of reforms, such as public-private partnerships in delivering public services and the use of new management techniques such as "Strategic Management," a substantial portion of these reforms are targeted at

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increasing the responsiveness of government to its "customers." As Osborne and Gaebler (1992, p. 169) explain, " . . . entrepreneurial governments have begun to transform themselves. They have begun to listen carefully to their customers, through customer surveys, focus groups, and a wide variety of other methods. . . . "

The task of using citizens' evaluations of the quality of service provision as an accountability mechanism, however, may be more complex than is commonly understood. For instance, many studies of citizens' satisfaction with services simply ask citizens to either assign a letter grade to the quality of services or rate their experiences from "very satisfying" to "very dissatisfying" (e.g., DeHoog, Lowery, and Lyons 1990; Beck, Rainey, Traut 1990). These studies, however, not only fail to distinguish among measures of service quality and satisfaction, but also to consider citizens' expectations about the quality of services. Work by scholars such as Kahneman and Tverksy (1979, 1982) in the psychology of decision-making has demonstrated the importance of a shifting reference point. In this article, we examine the relationship between citizens' assessments of services with respect to a reference point and citizens' satisfaction with governmental services.

We begin by examining past research that has explored the underlying determinants of citizen satisfaction with services. Next, we focus specifically on the role of expectations and perceived performance in determining satisfaction. We then examine empirically, across three policy areas, the relationship between expectations, perceived service quality, and satisfaction. We consider the significance of our findings in the article's conclusion.

WHAT DRIVES CITIZENS' SATISFACTION WITH SERVICES?

While the subject of a significant volume of research in 1970s and early 1980s (e.g. Hirschman 1970; Ostrom 1976; Lineberry 1977; Sharp 1986), it is only in recent years that citizens' satisfaction with local services has begun to attract renewed scholarly attention. Understanding citizens' levels of satisfaction with services provides an important assessment mechanism for understanding the level of performance of governmental agencies (in response to customer-oriented service reforms) as well as the impact of changes in the institutional arrangements governing the delivery of services, such as choice in the area of education or the use of nonprofits in the delivery of social services (see e.g. Van Slyke and Roch 2004).

When considering what drives citizens' assessments of their experiences with services, many service-satisfaction studies have investigated the rela-

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tionship between demographic traits and satisfaction. At a minimum, there are at least three different ways to view this relationship (see DeHoog, Lowery, and Lyons 1990). One possible explanation is that these characteristics simply reflect variations in the supply of services across jurisdictions. Thus, in many of these studies, satisfaction is used as a proxy indicator of service quality. For instance, Blacks have been consistently found to be less satisfied with services (Jacob 1972; Durand 1976; Brown and Coulter 1983; but see also Lineberry 1977; Jones 1980), suggesting, according to some scholars, that jurisdictions with larger numbers of Black residents provide lower-quality services. A second possible explanation is that demographic characteristics reflect different levels of demand for different services found in neighborhoods of varying status (Sharp 1986), and a greater demand for services may be more difficult for service providers to satisfy. And finally, differences in satisfaction among individuals may exist because differences in demographic characteristics—such as gender, age, and income—are linked to varying political attitudes that are closely related to satisfaction. For instance, local political efficacy has been found to be positively related to service satisfaction in the area of police services (Brown and Coulter 1983), as well as to satisfaction with services more generally using the closely related concepts of *community disaffection* (Stipak 1979; Beck et al. 1990) and *community attachment* (Lyons and Lowery 1989).

Other studies have focused less on demographic characteristics and more so on the relationship between the quality of the services provided and satisfaction. When considering this relationship, one key issue for scholars has been whether changes in levels of satisfaction with services correspond to changes in objective performance measures. For instance, research suggests that citizens may disregard service characteristics that influence more objective measures of quality. Stipak (1977, p. 50) found that "respondents were largely ignorant of the basic facts about how their local services were provided." Researchers have also consistently documented a gap between citizens' levels of perceived quality of governmental services and objective measures of program performance. Link and Oldendick (2000, p. 142) found that "perceptions [of service quality were] rarely correlate[d] with objective output measures." As a result, much research has focused on the role of perceived quality, rather than objective quality, as a driver of service satisfaction (e.g. Serra 1995; Van Ryzin 2004, but see DeHoog, Lowery, and Lyons 1990).

Thus, when considering citizens' satisfaction with services, previous research has examined the influence of factors such as demographic characteristics and varying levels of service quality. How do citizens' expectations of the quality of services, however, relate to service satisfaction?

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THE ROLE OF EXPECTATIONS IN SERVICE SATISFACTION

Much previous research investigating the relationship between expectations and service satisfaction has focused on private-sector services. For instance, in the field of marketing, expectations are widely regarded as playing a key role in determining consumers' levels of satisfaction with goods and services. Much of this research is grounded in the disconfirmation paradigm, defined by Oliver (1977) and based on the notion that satisfaction is closely linked to consumers' perceptions of whether experiences with a product or service positively or negatively disconfirm their prepurchase expectations (Anderson 1973; Oliver 1980, 1997). This body of research clearly identifies expectations and perceived quality as constructs separate from service satisfaction and recognizes these as two key antecedents of satisfaction (Johnson and Fornell 1991; Anderson and Sullivan 1993). Findings from this body of work also suggest that the influence of expectations may vary across market sectors. The influence of expectations appears greater in sectors in which standardization makes it easier for consumers to assess service quality and in which consumers make more frequent consumption decisions (Fornell et al. 1996). In addition, as predicted by Kahneman and Tversky's prospect theory, researchers have also found the influence of the negative disconfirmation of expectations on satisfaction to be greater than that of positive disconfirmation (Kopalle and Lehmann 2001; Kahneman and Tversky 1982). Overall, a significant body of consumer research demonstrates that the relationship between the perceived quality of the service and an initial reference point is critical to understanding consumers' reported levels of satisfaction with services.

The notion that satisfaction is based in part on the positive disconfirmation of expectations is also evident in work in political science focused on ratings of political elites. For instance, in their work Kimball and Patterson (1997) demonstrate empirically that the "expectations-perception discrepancy" negatively influences Americans' evaluations of Congress. Waterman, Jenkins-Smith, and Silva (1999, p. 945) hypothesize a similar relationship between expectations and assessments of presidential performance ". . . expectations are seen as shaping evaluations. Since expectations are excessive and contradictory, evaluations of presidential performance are negatively affected, which then constrains presidential performance . . ." Their empirical investigation similarly reveals that a larger gap between expectations and performance is associated with lower levels of presidential job approval and decreases the likelihood that a citizen will vote for an incumbent president.

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We know, however, relatively little about the relationship between disconfirmation of expectations and satisfaction with public-sector services. The disconfirmation of expectations is a central component of the SERVQUAL model developed in the private sector (Parasuraman, Zeithaml, and Berry 1988; Parasuraman, Berry, and Zeithaml 1991) and applied in some recent research to the area of public-sector services; for example, Wisniewski and Donnelly (1996) use the model in their study of public library users in the United Kingdom. This model, however, assesses the disconfirmation of expectations across different dimensions of a product or service in order to establish a global measure of service quality; it does not explicitly link disconfirmation and satisfaction. The link between disconfirmation and satisfaction is evident in recent research by Van Ryzin (2004) that investigates the influence of the disconfirmation of expectations on citizens' overall satisfaction with New York City services. The results of this investigation indicate that disconfirmation is positively related to satisfaction. It is important to note, however, that it is not possible from these results to determine whether the effects of disconfirmed expectations vary across different service areas. Furthermore, these results may in fact understate the influence of disconfirmation, because they are based on a substitute rather than direct measure of disconfirmation, as discussed by Van Ryzin (2004, p. 440): "[I]t should be noted that there is evidence in the consumer behavior literature of the advantage of asking an explicit question about disconfirmation . . . because such variables have been found to better explain variance in satisfaction." Examining business executives' satisfaction with business policy in Norway, Adreassen (1994) has also investigated the role of disconfirmation of expectations and perceived service quality as two key drivers of satisfaction. This work is, however, based on a measure that fails to distinguish whether experiences in fact positively or negatively disconfirm an initial set of expectations.

Thus, while a small body of previous work has examined the relationship between the disconfirmation of expectations and satisfaction with public services, much of this research is based on the use of unsatisfactory measures and focused on overall satisfaction with a set of services or on a single service area.

METHOD

We base our empirical study on data from a statewide telephone survey of adults 18 and older who live in Georgia, called the *Georgia Poll*, conducted by the Applied Research Center at Georgia State University during spring 2002. For the spring 2002 poll, 744 residents were randomly selected and

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TABLE 1: The Expectations-Performance Differential Across Service Areas

Performance Expectations	*Trash*	*Police*	*Schools*
Performance exceeds expectations	31.03%	16.95%	19.90%
Performance meets expectations	55.97%	62.99%	49.25%
Performance falls short of expectations	13.00%	20.06%	30.85%
N	477	643	201

SOURCE: Georgia State Poll.

interviewed from April 18 to June 14, 2002, on a variety of public policy issues that included citizens' experiences with local services. The response rate was 29%.[1] All adults living in Georgia with a working telephone (including new and unlisted numbers) were eligible for the poll.

RESULTS

We begin by simply examining how the gap between expectations and performance varies across our three different service areas. Survey respondents were asked "How does the overall performance of [ ] stack up against your expectations? Would you say overall performance exceeds expectations, meets your expectations, falls short of your expectations?" Their responses to this question are summarized in Table 1.

The modal response in all three cases is "meets expectations." Note, however, that smaller percentages of respondents are likely to report that the performance of the police or the schools exceeds expectations. Thus, in the case of trash services, it appears that respondents are either more likely to possess low expectations, which are easier to exceed, or that respondents perceive the quality of their trash services as higher than that of the police or the local schools.

In the survey, respondents were also asked to evaluate the perceived quality of these different services. Before respondents were asked how performance stacks up against expectations, they were asked "In general how would you rate the overall performance of [ ]? Use the same scale that schools use to grade students' performance, where an "A" means excellent, a "B" means good, a "C" means fair, a "D" means poor, and "F" means failing. What grade would you give for the overall performance of [ ]?" While this letter-grade scale has been most commonly used when asking parents to assess the quality of their children's schools, it is increasingly being used to gauge citizens' assessments of a broader range of local services (see e.g. Harrington 2005; Hembroff and Sierra 1998). This scale offers a familiar metric,

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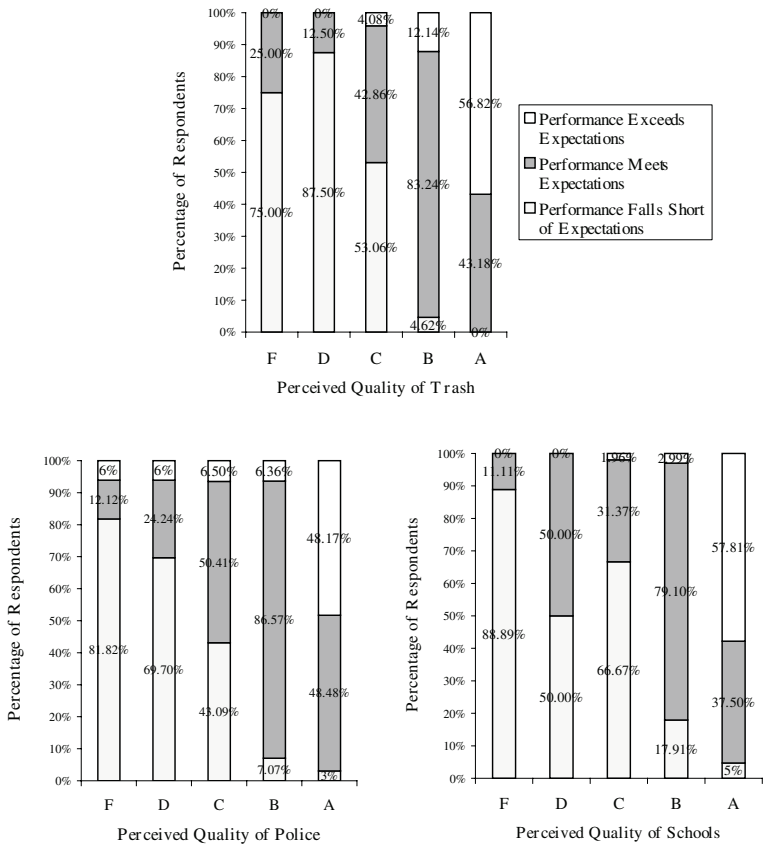


Figure 1: The Relationship Between the Perceived Quality of Services and the Disconfirmation of Expectations

and when translated into the common 4-point scale used by many academic institutions, it is easy to interpret across a range of values. In response to this question, most respondents reported that they perceived the quality of services as either an "A" or "B"; a substantial percentage, however, also rated the quality of a service as a "C" or lower (16% [trash]; 29% [police]; 34% [schools]). Note that these percentages suggest that respondents perceive their trash services more positively than their police or local schools.

In Figure 1, we examine the relationship between the disconfirmation of expectations and the perceived quality of services. Overall, higher-quality services appear more likely to exceed expectations. The standard, however,

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by which respondents evaluate these different services clearly varies; for instance, an "A," in some cases, only meets rather than exceeds expectations. Thus, in these cases, some respondents appear to have even higher expectations; perhaps only an "A+" service is likely to exceed them. In all three cases, a service rating a "B" is likely to meet expectations, an "F" is likely to fall short of expectations, and there exists more substantial variation in the relationship between "C" and "D" ratings and citizens' expectations.

Thus, this first cut at our data suggests that an "A" service is not always evaluated according to the same reference point. In some cases, services that are perceived as high in quality exceed citizen-consumer expectations, and in other cases they simply meet expectations.

Examining Citizens' Satisfaction with Services

How do perceived quality and expectations relate to satisfaction? Are citizen-consumers more likely to be satisfied with services that exceed their expectations? Does the influence of the disconfirmation of expectations vary across different service areas? And is the influence of the negative disconfirmation of expectations greater than that of the positive disconfirmation? We investigate these questions by estimating a series of ordered probit models that take on the following form:

Satisfaction with Services = f(perceived quality of service, performance-expectations differential, demographic factors, attachment to the community, and controls)

In the survey, at the end of each set of questions about each particular service area, respondents were asked "Overall, to what extent are you satisfied with [service name]? Are you very satisfied, satisfied, neutral, dissatisfied, very dissatisfied?" We use the responses to this question as a dependent variable in each of our models.

We expect that satisfaction will be influenced by perceived quality, which is the same measure that we present in Figure 1, as well as by the disconfirmation of expectations. In order to assess the impact of expectations, we include two dummy variables that indicate whether the respondents assessed performance as either exceeding expectations or falling short of expectations. Thus, meeting expectations is the omitted category. These dummy variables allow us to investigate whether falling short of expectations has a larger effect on the reported levels of satisfaction than exceeding expectations.

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We also include in our models factors identified by previous research as likely to influence satisfaction with services. Thus, we consider the influence of demographic characteristics—such as race, education, and age—that have been found to influence citizen-consumers' satisfaction with providers. For instance, we expect that Blacks will be less satisfied with services than non-Blacks (Durand 1976; Brown and Coulter 1983). Age, which we include coded in years, may also influence satisfaction to the extent that this factor is associated with a greater demand for services (which may be more difficult to satisfy). For instance, more elderly clients are likely to have higher levels of fear of crime (Warr 1984) and thus may make greater demands of police. Younger citizen-consumers, on the other hand, should demand more of the local schools. In the case of trash services, age most likely has little effect but is included to increase uniformity across models. Education, which we include as a dummy variable that takes on a value of 1 when the respondent has attended college, may act as a proxy for local political involvement and thus we expect that higher-educated respondents should have higher levels of service satisfaction.

We also consider the local political efficacy of the respondent directly. We rely on two dummy variables—owning a home and whether the respondent reported being a registered voter. Local political involvement should signify greater attachment to the community, and thus we expect that these factors should be associated with higher service expectations. As other controls, we include a variable that indicates whether the respondent lives in the metro Atlanta area. In the case of schools, we include an additional variable that indicates whether the respondents' children attend a private school.

We estimate our models using a probit estimation procedure that provides pseudo-maximum likelihood estimates and Huber standard errors. (See Skinner 1989 for a discussion of pseudo-maximum likelihood estimates.) We employ this procedure rather than the standard probit procedure to account for the clustering of respondents in particular county areas, where respondents are likely to rate the same services. Thus, these standard errors have been adjusted to take into account dependence among observations in defined clusters—where each cluster includes all the respondents that reside in a particular county (Rogers 1993; see also Huber 1967).

We present the results of our analyses in Table 2. The results indicate that exceeding expectations leads to higher levels of satisfaction, while falling short of expectations leads to lower levels of satisfaction. Perceived service quality is also significantly associated with higher levels of satisfaction. The magnitude of these effects does not appear to vary greatly across our three chosen service areas. Furthermore, our results do not indicate that the negative disconfirmation of expectations has a larger influence on satisfaction

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TABLE 2: Factors Influencing the Levels of Satisfaction Across Three Different Service Areas

	Trash		*Policing*		*Schools*	
	Coefficient	*Standard Error*	*Coefficient*	*Standard Error*	*Coefficient*	*Standard Error*
Exceeds expectations	1.202***	.168	1.058***	.157	1.029**	.325
Fall shorts of expectations	–.895***	.212	–.708***	.172	–1.103***	.251
Grade given provider	.716***	.103	.658***	.064	.629***	.141
Education	–.103	.120	.105	.100	–.110	.149
Black	–.161	.126	–.159**	.082	–.081	.160
Other	–.319	.157	.093	.230	–.237	.255
Age	.002	.003	.001	.003	–.002	.009
Own home	–.055	.136	–.068	.162	–.087	.211
Vote	.105	.138	.107	.123	–.132	.318
Private school					.281	.205
Metro area	–.006	.105	–.034	.089	–.023	.137
Cut 1	–.073		–.182		–.784	
Cut 2	.841		.788		.417	
Cut 3	1.881		1.757		1.334	
Cut 4	4.078		3.954		3.442	
χ^2	268.19***		357.57***		216.51***	
N	439		587		183	

SOURCE: Georgia State Poll

NOTE: Cut 1 and so on refer to cut-points on a standardized normal distribution that are used to calculate predicted probabilities for each category of the dependent variable.

*$p < .1$. **$p < .05$. *** $p < .01$. Two-tailed test.

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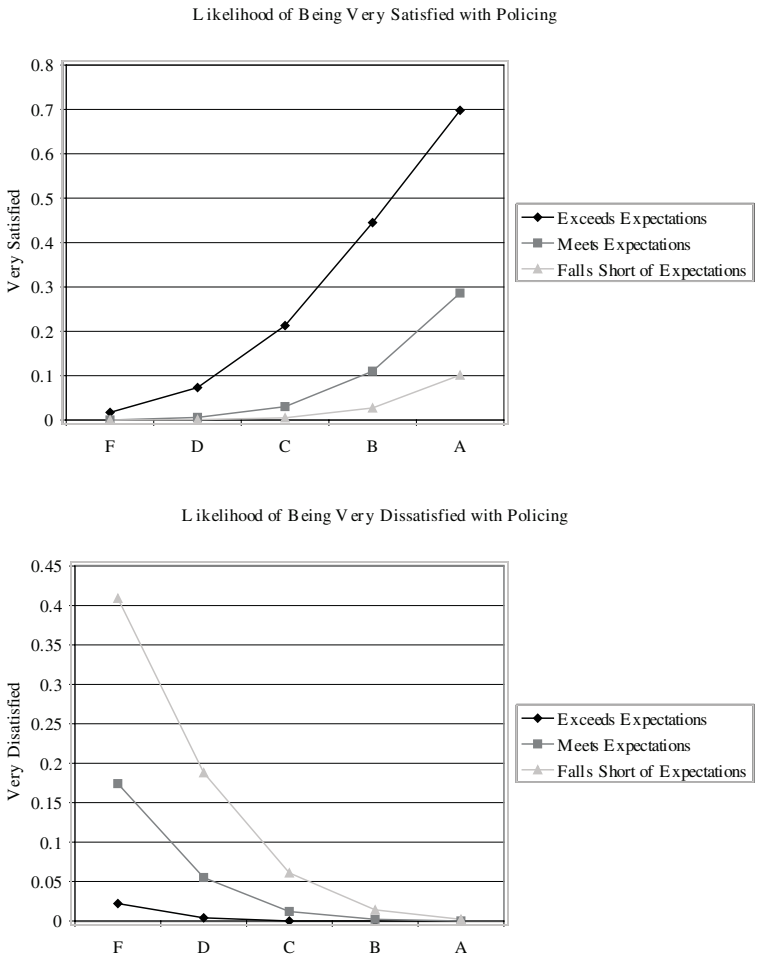


Figure 2: The Influence of Expectations and Perceived Performance on Satisfaction with Police

than the positive disconfirmation of expectations. Note that Blacks are less likely than Whites to express satisfaction with the police.

We also calculate the predicted probabilities associated with changing values of our measure of expectations and service quality. Note that we hold

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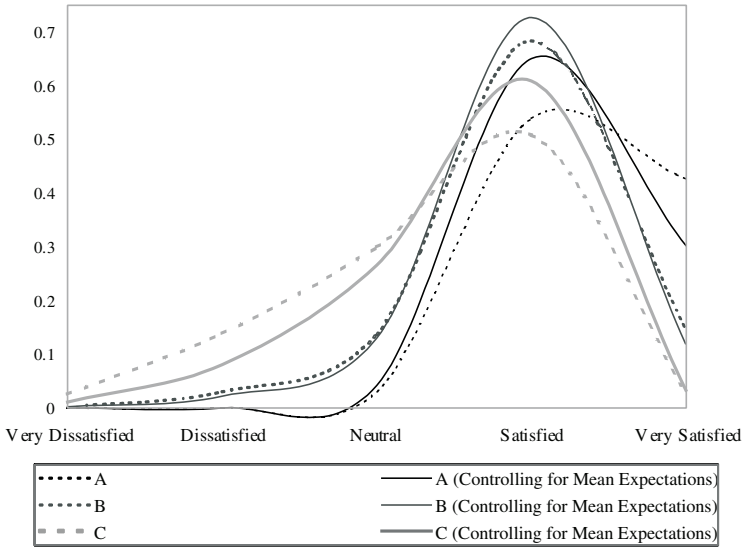


Figure 3: The Effect of Controlling for Expectations on the Relationship Between Perceived Service Quality and Predicted Levels of Satisfaction with Police

all dummy variables at the modal category and all other variables at their mean. Since the patterns we report are similar across our three service areas, we present only the predicted probabilities for satisfaction with policing in Figure 2.

Figure 2 demonstrates the dual influences of expectations and perceived service quality on satisfaction. When perceived quality of a service is high, say an "A" or "B" service, and performance exceeds expectations, citizen-consumers are more likely to be very satisfied than if performance meets or falls short of expectations. Thus, higher expectations appear to decrease the likelihood of falling into the highest satisfaction category. When perceived service quality is low, say a "D" or "F" service, and performance falls short of expectations, citizen-consumers are more likely to be very dissatisfied than if performance meets or exceeds expectations. Thus, when considering research that relies on satisfaction as an indicator of service quality, our results suggest that similar levels of satisfaction are likely to reflect varying levels of service quality as citizens' expectations about the quality of services vary.

We further illustrate the effect of disconfirmation on satisfaction in Figure 3. We first present the change in likelihood that a citizen will report different

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levels of service satisfaction, based on the models that we present in Table 2, and then compare that effect to the likelihood based on an additional set of models that omit the control for disconfirmation (model results are available in the appendix).[2] When generating these predicted probabilities, we hold our measure of disconfirmation constant at its mean (when included) and vary the values of perceived service quality from "A" to "C."

The results demonstrate that once expectations are taken into account, the distribution for satisfaction is somewhat attenuated and the likelihood of falling into the satisfied category increases. For instance, when considering satisfaction as a function of perceived service quality, held constant at a "C" value, the predicted likelihood of falling into the categories "dissatisfied" and "very dissatisfied" is greater than when satisfaction is modeled as a function of perceived service quality and the disconfirmation of expectations. The effect is also similar when we hold perceived service quality constant at a "B" or "A" value.

CONCLUSION

This research has explored the relationship between perceived service quality, expectations, and satisfaction across three service areas: trash services, police, and schools. It contributes to a growing body of research examining the role of expectations in citizen-consumers' levels of satisfaction with governmental services.

The results demonstrate the important role that the disconfirmation of expectations play in determining citizens' levels of satisfaction with services. Citizen-consumers report varying levels of satisfaction with services that they perceive to be of similar quality. Thus, an "A" service is not always associated with the same level of service satisfaction—the relationship between the perceived quality of a service and a pre-existing reference point figures prominently in determining citizen satisfaction with services. Our results also demonstrate the strong influence of perceived service quality on satisfaction with services. Services that are perceived as high in quality are clearly likely to lead to higher levels of satisfaction than those that are perceived poorly. High expectations and lower-quality services appear much less likely to lead to higher levels of citizen-consumer satisfaction.

Thus, this research emphasizes the importance of considering the disconfirmation of expectations, perceived service quality, and service satisfaction not simply as measures of a single underlying construct. Expectations and perceived service quality clearly have a significant and independent influence on satisfaction.

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This research also suggests that past research that relies on satisfaction as an indicator of perceived service quality may observe a greater variation in satisfaction that is in part due to the variation in expectations among citizens. Since the overall effect of expectations appears to attenuate the distribution of satisfaction, studies that rely on mean levels of satisfaction may not be subject to systematic error at this aggregate level. To the extent, however, that expectations are influenced by factors, such as institutional arrangements governing the delivery of services, that are of interest to scholars in this area, systematic variations in satisfaction may express differences in expectations rather than variations in service quality or in the match between service-tax bundle and citizen-consumer preferences. Thus, a central task for future research is to investigate carefully the factors that underlie the formation of citizens' expectations for public services.

While we believe that our direct measure of disconfirmation and focus on multiple service areas allows us to further previous research on expectations and satisfaction with public-sector services, we also note a couple of limitations of this current study. First, it is important to note that these analyses are based on the results of a survey of citizens in a single state; thus, findings may in fact differ across states and metropolitan areas. Second, our results are also limited to a subset of service areas; findings may well differ for other types of services or services offered at different levels of government.

In sum, this research attempts to contribute to our understanding of a key issue in democratic governance—how citizens form judgments about the quality of public services. These findings suggest that understanding the role of the disconfirmation of expectations is central to this process.

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APPENDIX
Factors Influencing the Levels of Satisfaction Across Three Different Service Areas—Omitting the Control for Expectations

	Trash		*Policing*		*Schools*	
	Coefficient	*Standard Error*	*Coefficient*	*Standard Error*	*Coefficient*	*Standard Error*
Grade given provider	1.075***	.107	.869***	.069	.968***	.129
Education	–.106	.146	.154	.104	–.065	.137
Black	–.157	.119	–.118	.079	.071	.172
Other	–.349*	.181	–.048	.238	–.106	.270
Age	.003	.003	.001	.003	–.004	.008
Own home	–.108	.173	–.021	.159	.062	.202
Vote	.150	.140	.028	.135	–.029	.261
Private school					–.095	.147
Metro area	–.059	.116	–.052	.090	.418	.192
Cut 1	1.385		.715		1.118	
Cut 2	2.354		1.687		2.274	
Cut 3	3.316		2.573		3.089	
Cut 4	5.298		4.583		4.891	
χ^2	241.08***		248.7***		89.38***	
N	443		595		185	

SOURCE: Georgia State Poll
NOTE: Cut 1 and so on refer to cut-points on a standardized normal distribution that are used to calculate predicted probabilities for each category of the dependent variable.
*$p < .1$. **$p < .05$. ***$p < .01$.
Two-tailed test.

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NOTES

1. This rate was calculated according to the formula AAPOR 3; the AAPOR 1 rate was 28% (see AAPOR Standard Definitions 2004).

2. We rely on the same estimation procedure as in our previous models. In the models in which we omit the control for disconfirmation, the coefficients for perceived service quality are about one-third larger than in the models that include the control. This difference suggests that the influence of perceived service quality on satisfaction occurs in part through the disconfirmation process.

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